DEL MAR
EST. SEAFOODS 1988

PROJECT BRIEF & TECHNICAL REQUIREMENTS · 项目简报与技术要求

VERTICAL PLATE FREEZER PROJECT

立式平板冻结项目 — 技术要求简报

Prepared by Del Mar Seafoods for Jiangsu Wolfking Tech Co., Ltd. and APCCO. English version followed by the Chinese version.
由 Del Mar Seafoods 编制，致江苏狼王智能科技股份有限公司及 APCCO。先英文版，后中文版。

DOCUMENT	DATE	SITE	STATUS
DMS-VPF-001 · Rev A	30 October 2026	Watsonville, California, USA	Issued for design start

PART 1 OF 2 — ENGLISH

1 PURPOSE

This document brings together everything Del Mar Seafoods has decided for its vertical plate freezer project in Watsonville, California, and sets out how Del Mar, Wolfking and APCCO divide the work. It is the starting point for Wolfking's design. It replaces nothing already agreed by email; it collects it in one place.

Each requirement carries a status so there is no doubt about who owns it:

CONFIRMED decided by Del Mar; fixed unless Del Mar changes it

WOLFKING Wolfking to design or propose

APCCO APCCO owns or will confirm

OPEN not yet decided; owner and timing listed in §12

> **HOW WE WORK**
>
> One shared email thread ("Del Mar plate freezing projects", started by Joe on 30 October 2026) for all project communication, with Del Mar, Wolfking and APCCO on every message. APCCO's engineers will ask many detailed questions. These are for the California permit, not a sign of problems; please answer them directly.

2 PARTIES AND ROLES

PARTY	ROLE IN THIS PROJECT
Del Mar Seafoods, Inc. Watsonville, CA — owner and operator	Buys the equipment, owns the site and the process. Joe Cappuccio (President): commercial decisions, product and schedule. Kevin Cappuccio: project engineering, electrical, technical liaison. Del Mar approves the block size, the layout and the final design before manufacturing.
Jiangsu Wolfking Tech Co., Ltd. Nantong, Jiangsu — designer and manufacturer	Designs and manufactures four vertical plate freezers with hydraulic power unit(s) and controls, and the complete ammonia refrigeration package for them (packaged engine room preferred, see §6). Supplies design drawings, calculations, certificates and documentation needed for the California permits. Sends two technicians to site to guide APCCO's crew during installation and to lead commissioning. Supports Del Mar's phase-2 automation (loading, unloading, bagging and boxing) with its partners.
APCCO — Applied Process Cooling Corp. Fresno, CA — licensed refrigeration contractor and engineer	Reviews Wolfking's design for California compliance, prepares and stamps permit drawings and calculations, files the permits with the City of Watsonville and other agencies. Installs all delivered equipment: rigging and setting, ammonia piping and insulation, electrical connections, controls integration. Performs start-up with Wolfking and provides ongoing maintenance and service. APCCO does not manufacture freezers; it is Del Mar's local partner for installation and compliance and will work directly with Wolfking's engineers.

CERTIFICATES, AGREED UP FRONT

To avoid the problem Wolfking described from a previous export (documents demanded at the port after shipment), APCCO will give Wolfking a written list of every certificate, test report and document required for this installation before manufacturing starts, and Wolfking will confirm which it can provide. APCCO

3 PRODUCTS AND PRODUCTION TARGETS

TWO PRODUCTS ON THE SAME FREEZERS CONFIRMED

- **Whole California market squid** (*Doryteuthis opalescens*), plate-frozen in blocks. This is Del Mar's test product for the China market: if Chinese buyers accept plate-frozen squid, Del Mar will convert more production to this method. The Monterey Bay squid season starts in May, so the line must be running by May 2027.
- **Pet-food raw material**: ground squid, a paste-like mince, frozen in the same blocks. This runs year-round under an existing contract and is the base load that pays for the equipment.

DAILY PRODUCTION CONFIRMED

45 t	2 × 10 h	4	8	140
per day (50 short tons), pet-food contract	shifts per day + 4 h sanitation	vertical plate freezers, identical	batches per freezer per day (4 per shift)	blocks of 10 kg per batch (target)

Del Mar's cycle assumption: 90 minutes freezing plus 30 minutes to unload and reload, so one batch every 2 hours and four batches per shift. 45 t ÷ 8 batches ÷ 4 freezers gives about 1.4 t, or 140 blocks of 10 kg, per freezer per batch. 140 blocks is the goal; fewer is acceptable if the design cannot reach it, but please try.

BACKGROUND AND WHAT COMES NEXT

Today Del Mar freezes large volumes of squid in cartons on racks in blast freezers. That is very labor-intensive at California labor rates. The goal of this project is block freezing with far less handling. The four freezers are a pilot: if the product and the economics work, Del Mar will add freezers in multiples of this design, so Wolfking should design for repeatability. Phase 2 (planned for 2027–2028) is automation: automatic loading of the freezers, automatic unloading, bagging and carton packing with a check-weigher. The phase-1 freezers must be ready to accept that automation later (see §5).

4 BLOCK SPECIFICATION

ITEM	REQUIREMENT	STATUS
Block mass	10 kg nominal. Squid is about 20% less dense than water, so a 12-litre block gives roughly 10 kg (10 kg of water = 10 L).	CONFIRMED
Block volume	12,000,000 mm^3 (12.0 L) nominal.	CONFIRMED
Thickness	75 mm. Fixed; this sets the freezing time.	CONFIRMED
Face size	Target about 535 × 300 mm (535 × 300 × 75 = 12.04 L). Del Mar is flexible on the exact face dimensions, but whatever is chosen becomes the Del Mar standard block for all future freezers. Wolfking to propose the face size that best fits its plate extrusions and still gives 12 L; Del Mar approves before plates are ordered.	WOLFKING
Blocks per station	Working assumption from the Qingdao meeting: two divider plates per station, three blocks side by side. Wolfking to confirm against the chosen face size, plate height and block orientation (which dimension is vertical).	WOLFKING
Blocks per batch	140 per freezer is the target (117–120 was discussed at the show; Joe's email of 30 October raises it to 140). Design as close to 140 as practical and state the resulting plate count and frame length.	WOLFKING
Freezing time	Del Mar assumes 90 minutes for a 75 mm block. Wolfking to confirm the time to −18 °C core at the design suction temperature, for whole squid and for ground squid paste, and to state the assumptions (product in-feed temperature, plate temperature, contact).	WOLFKING
Drainage of free liquid	Whole squid releases a lot of water and ink when loaded and in the first minutes of freezing. The bottom of each station must let this free liquid drain away rather than trap it, so the customer receives a block of squid, not a block of ice. But the pet-food paste must not run out of the station. Wolfking to propose a bottom design that handles both (for example a drain gap or holes sized to pass thin liquid but hold paste, or an adjustable arrangement). Del Mar will send product photos or samples if useful.	WOLFKING

5 VERTICAL PLATE FREEZER REQUIREMENTS

ITEM	REQUIREMENT	STATUS
Quantity and type	Four identical vertical plate freezers, top-loaded, installed indoors in the processing room. Hydraulic plate closing/opening, hot-gas release and hydraulic block lift. Block discharge to one side of the freezer onto a table or conveyor; the discharge side is fixed in the layout (§8).	CONFIRMED
Capacity	Per §3 and §4: 75 mm blocks, 3 per station, 140 per batch target, 8 batches per day.	CONFIRMED
Loading, phase 1	Manual or semi-automatic. Del Mar is considering a top conveyor with gates for whole squid and hose/pump filling for paste; Wolfking's advice on the simplest reliable method for 140 stations is welcome. Final choice is Del Mar's.	OPEN
Ready for phase-2 automation	The freezers must accept Wolfking's automatic loading and unloading system later without being replaced. Design in the interface now: top clearance, mounting points or rails for a loading gantry, the block discharge conveyor interface, and spare control I/O. Wolfking to state what provisions it needs built in.	WOLFKING
Unloading and packing	After the lift, blocks go to bagging and carton packing (block in a bag, bag in a box), with a check-weigher planned for phase 2. Wolfking offered a partner's bagging and case-sealing machine; please include it as a separately priced option.	WOLFKING
Materials and hygiene	Plates: aluminium alloy (extruded), food-contact grade. Product zone and splash zone, including everything above the plates: stainless steel or aluminium, cleanable, no galvanized surfaces. Structural base below the product zone: Wolfking to propose stainless steel or hot-dip galvanized with the price difference; the plant handles salt water and chlorinated sanitizers daily.	WOLFKING
Hydraulic power unit	Motor 460 V / 60 Hz (§7). Located outside the cold and wet zone. Oil and seals compatible with NSF H1 food-grade hydraulic oil, and rated for −35 °C at the cylinders and hoses. Wolfking to state motor kW, working pressure, and whether it proposes one unit per freezer or one shared unit.	WOLFKING
Refrigerant connections	Flexible connections to the moving plates: stainless steel, rated for ammonia, replaceable from outside. No copper or brass anywhere in the ammonia circuit.	CONFIRMED

ITEM	REQUIREMENT	STATUS
Safety	Emergency stop, guarding of the pinch points at the open top, lockable electrical disconnect, means to isolate and bleed hydraulic energy. Guarding to US practice; APCCO will review (§9).	APCCO
Documentation	In English: general arrangement drawings, P&ID, electrical schematics, operation and maintenance manual, spare-parts list with a recommended two-year spares package (hoses, seals, valves). 3D models (STEP) of the freezer and skids for Del Mar's layout work.	CONFIRMED

6 REFRIGERATION PACKAGE

Wolfking designs and supplies the complete refrigeration system for the four freezers. APCCO reviews it for California compliance, installs it and maintains it.

ITEM	REQUIREMENT	STATUS
Refrigerant	Anhydrous ammonia, R717. California no longer allows high-GWP HFC refrigerants in new systems of this size, so ammonia is the practical choice.	CONFIRMED
Scope	Compressor package, condenser, vessels, ammonia pumps if a recirculated design, valve stations, safety reliefs, controls and control panel, and the interconnecting piping design. APCCO installs the field piping, insulation and electrical.	CONFIRMED
Arrangement	Del Mar's preference, discussed in Qingdao: a packaged or containerized engine room located outdoors, next to the building, with the freezers indoors and insulated piping between them. This saves building space and keeps compressor noise outside. Del Mar and APCCO meet at the Watsonville site in early November to settle the location and the piping route; Wolfking to send its standard packaged engine room dimensions and a concept layout before then. Alternative Joe asked about: a small self-contained refrigeration unit on each freezer; Wolfking to comment on feasibility, charge per unit and cost compared with one central package.	OPEN

ITEM	REQUIREMENT	STATUS
Ammonia charge	Minimum charge is a design goal. California's CalARP program applies at 500 lb (227 kg) of ammonia on site, and staying below that line simplifies permitting considerably. Wolfking to calculate the total charge for its proposed design. Del Mar understands it may not be possible at this capacity; if not, state the charge so APCCO can plan the filing.	WOLFKING
Design conditions	APCCO's early estimate, made before the batch size was raised to 140 blocks: about 45 TR (158 kW) at −30 °F (−34 °C) suction for four freezers, pumped recirculation assumed at 4:1. Wolfking to provide its own load calculation for the final batch size, including the peak pull-down load with staggered batches, and the resulting compressor and condenser selection.	WOLFKING
System type	Wolfking to state: pumped recirculation or other feed method and the recirculation ratio; suction and condensing temperatures; condenser type (evaporative or air-cooled; water use and noise both matter at this site); receiver or critical-charge design; defrost method (hot gas assumed) and the hot-gas requirements of the freezers.	WOLFKING
Valves and controls	Wolfking to provide a valve and device schedule with brands, and to state which valves ship mounted on the freezers, which ship loose, and which it expects APCCO to supply. State whether the freezer control panel controls the refrigeration valves. APCCO will confirm which brands are acceptable for service in California.	WOLFKING
Pressure design	Pressure vessels ASME Section VIII Division 1 with U-stamp and National Board registration. Piping to ASME B31.5. Design pressures per IIAR 2; the plates and their piping see discharge-side pressure during hot-gas defrost, so they must be rated for it (APCCO will confirm the required values, typically 250 psig / 17.2 bar). Vessel welds radiographed where the California vessel code requires it. Material certificates and welder qualifications available on request.	APCCO

7 ELECTRICAL

ITEM	REQUIREMENT	STATUS
Site power	480 V, 3-phase, 60 Hz (480Y/277 V, grounded wye). All motors nameplated 460 V / 60 Hz. There is no 380 V and no 220 V single-phase supply in the United States.	CONFIRMED

ITEM	REQUIREMENT	STATUS
Control power	120 VAC, 60 Hz, single-phase, from a control transformer inside each panel; 24 VDC for instruments, I/O and solenoid valves. No 220 V single-phase components (heaters, fans, lights, coils).	CONFIRMED
Control panels	Built and labelled to UL 508A (a UL or ETL listing to UL 508A), with the short-circuit current rating marked. NEMA 4X stainless enclosures in washdown areas. Components UL-listed or UL-recognized (Schneider, ABB, Siemens, Eaton, Allen-Bradley or equivalent). Wiring practice and colour code to NFPA 79; labels in English.	CONFIRMED
Motors	NEMA Premium / IE3 efficiency, which US Department of Energy rules require for imported motors even when they are built into a machine. Hydraulic pump and fan motors sized for 60 Hz speed, not relabelled 50 Hz units.	CONFIRMED
Load list, needed first	Per freezer and per skid: kW, full-load amps at 480 V, starting method and inrush, and the proposed number and size of feeders. Del Mar needs this to apply to the utility (PG&E) for service, which can take longer than manufacturing.	WOLFKING
Controls	PLC and HMI with English text. Remote access for Wolfking support is welcome; Del Mar provides the internet connection. Wolfking to propose the PLC brand; Siemens or Allen-Bradley preferred for service in the US.	WOLFKING

8 SITE: WATSONVILLE, CALIFORNIA

- **Location.** Del Mar Seafoods, Watsonville, California, on Monterey Bay about 150 km south of San Francisco. Mild coastal climate. APCCO will supply the design ambient temperatures for the condenser (summer design conditions are roughly 29 °C dry bulb / 18 °C wet bulb; to be confirmed). APCCO
- **Layout.** Freezers indoors in a processing room; packaged engine room outdoors adjacent to the building; piping route and lengths set at the Del Mar–APCCO site meeting in early November. Wolfking to state its preferred maximum distance between engine room and freezers. OPEN
- **Seismic.** Watsonville is in a high-seismic zone (the 1989 Loma Prieta earthquake was centred nearby). Every piece of equipment needs anchorage designed for California seismic loads. Wolfking to provide weights, centres of gravity and anchor-bolt locations for the freezers, skids and vessels; APCCO's engineer does the anchorage calculations. WOLFKING
- **Floor and foundations.** Wolfking to provide footprints and point loads for the freezers and the engine room package. WOLFKING

- **Utilities available.** 480 V 3-phase power, potable water, drains, internet. Compressed air to be confirmed.
- **Transport.** Ocean freight to the Port of Oakland or Long Beach, then road to Watsonville. A packaged engine room should fit standard container or legal road dimensions; APCCO will advise on rigging access at the site. Shipping terms to be agreed in the proposal. OPEN

9 REGULATORY NOTES FOR THE DESIGN PHASE

Only the points that change what Wolfking designs. APCCO is the authority on all of them and will give the specific values; these notes are so nothing is a surprise.

1. **Permits.** The City of Watsonville issues building, mechanical and electrical permits on drawings and calculations stamped by a California-licensed professional engineer (APCCO's P.E.). Wolfking supplies the design; APCCO adapts, stamps and files. Permits can take longer than manufacturing, which is why drawings come first.
2. **Ammonia system design.** ANSI/IIAR 2-2021 (design of closed-circuit ammonia systems) with IIAR 4, 5 and 6 for installation, start-up and inspection, plus the refrigeration chapter of the California Mechanical Code. These set design pressures, relief valves, valve arrangement and labelling, emergency shutdown, and ammonia detection and ventilation for any indoor machinery room. An outdoor packaged engine room reduces the indoor requirements.
3. **Ammonia inventory thresholds.** California's CalARP program applies from 500 lb (227 kg); federal OSHA PSM and EPA RMP from 10,000 lb. Design for minimum charge and document it.
4. **Vessels and piping.** ASME Section VIII Div. 1 U-stamp and National Board registration for vessels; ASME B31.5 for refrigeration piping; Cal/OSHA pressure-vessel permits, with radiography of vessel welds where required. Material certificates (MTRs) and welder qualifications to ASME IX must be available.
5. **Electrical.** California Electrical Code (NEC). Every piece of electrical equipment must be listed by a US testing laboratory (UL, ETL, CSA): panels to UL 508A, motors and components listed or recognized, motors to DOE efficiency levels. Unlisted equipment needs a field evaluation, which is slow and expensive, so design it out.
6. **Refrigerant.** California (CARB) prohibits high-GWP HFCs in new refrigeration systems with more than 50 lb charge; ammonia is the compliant choice and is already decided.
7. **Seismic.** California Building Code and ASCE 7: equipment anchorage and restraint designed for the site. Skids, vessels and freezer frames need designed anchor points.
8. **Food safety.** FDA FSMA: seafood HACCP (21 CFR 123) and animal-food preventive controls (21 CFR 507). Product-contact surfaces must be corrosion-resistant and cleanable; lubricants that could reach product must be NSF H1; no flaking coatings above product.
9. **Worker safety.** Cal/OSHA: machine guarding, emergency stops, lockout/tagout provisions, English labels and warnings.
10. **Noise.** Outdoor equipment next to the building must meet the city's noise limits at the property line. This affects condenser fan selection and whether the package needs acoustic treatment. Del Mar and APCCO will give the limit once the location is fixed.

10 SCHEDULE

WHEN	WHAT	WHO
Week of 2 Nov 2026	This brief issued. Del Mar sends product photos and any samples. Del Mar and APCCO meet at the Watsonville site on the engine room location and layout. Wolfking starts design.	All
As soon as possible	Preliminary general arrangement, P&ID, electrical load list, ammonia charge estimate and proposal (§11). Del Mar and APCCO review; permit work starts on the preliminary design.	Wolfking
After design approval	Manufacturing of freezers and refrigeration package. Wolfking's estimate at the show: about 100 days.	Wolfking
Shipping and installation	Ocean transit, customs, rigging, installation, commissioning. APCCO to confirm installation duration and crew.	APCCO + Wolfking technicians
May 2027	Target: installed and commissioned for the start of the squid season. June 2027 is the latest acceptable date. Joe has told the pet-food customer six months in total.	All
2027–2028	Phase 2: automatic loading and unloading, bagging and case packing, check-weighing.	Wolfking + partners

11 WHAT WE NEED FROM WOLFKING FIRST

1. **Preliminary general arrangement of one freezer:** dimensions, plate count, station layout, proposed block face size, loading height, discharge side, weights.
2. **Freezing-time calculation** for 75 mm blocks of whole squid and of squid paste, with assumptions.
3. **Refrigeration concept:** system type, P&ID, equipment list, ammonia charge estimate, packaged engine room concept with dimensions and weight.
4. **Electrical load list** and proposed one-line diagram; control-panel specification and the listing approach (UL or ETL).
5. **Vessel list** with ASME and National Board status; welding and QA documentation plan; list of certificates Wolfking can supply (to be matched against APCCO's list, §2).
6. **Commercial proposal:** scope split between Wolfking supply and APCCO installation, price, delivery, payment terms, warranty, recommended spares package, and terms for the on-site technicians (number of people, days, travel).
7. **3D models** (STEP) of the freezer and engine room package when available, for Del Mar's layout.
8. **Reference list** of US installations, including the Oregon and Los Angeles customers discussed in Qingdao.

9. **Phase-2 budget concept** for automatic loading and unloading and for the partner bagging and case-packing equipment, so the interfaces are designed into the freezers now.

12 OPEN ITEMS

ITEM	OWNER	NEEDED BY
Final block face dimensions (around 535 × 300 × 75 mm, 12 L)	Wolfking proposes, Del Mar approves	Before plates are ordered
Blocks per batch (140 target) and plate count	Wolfking	With preliminary GA
Freezing time confirmation	Wolfking	With preliminary GA
Station bottom design: drain whole-squid liquid, hold paste	Wolfking	With preliminary GA
Frame base material (stainless or hot-dip galvanized) and price difference	Wolfking	With proposal
One hydraulic unit per freezer or one shared	Wolfking	With preliminary GA
Engine room: containerized vs. open skid, location, noise treatment	Del Mar + APCCO site meeting	Early November
Ammonia charge and CalARP position	Wolfking calculates, APCCO reviews	With refrigeration concept
Condenser type and water availability	Wolfking proposes, APCCO confirms	With refrigeration concept
Hot-gas defrost requirements and valve scope split	Wolfking + APCCO	With P&ID
Electrical load list for the PG&E application	Wolfking	As soon as possible
Phase-1 loading method	Del Mar decides, Wolfking advises	With GA
Shipping terms and port	Del Mar + Wolfking	With proposal
Installation duration and crew size	APCCO	After preliminary design

13 CONTACTS

DEL MAR SEAFOODS

Joe Cappuccio
President
joe@delmarseafoods.com
+1 831-760-3005

DEL MAR SEAFOODS

Kevin Cappuccio
Project engineering, electrical
arabidsquid@gmail.com

APCCO

Alex Robinson
Division Manager, Construction
arobinson@apcco.net
+1 559-875-7720 ext. 411 · cell +1 559-696-6227

APCCO

Patrick Cocagne, P.E.
Vice President of Engineering
pcocagne@apcco.net
+1 559-875-7720 ext. 402 · cell +1 559-358-5063

WOLFKING

Sean Zhu
Jiangsu Wolfking Tech Co., Ltd.
sean@wolfkingtech.com

WOLFKING

David Ye
Jiangsu Wolfking Tech Co., Ltd.
david@wolfkingtech.com

Del Mar Seafoods, Inc. ·Watsonville, California ·www.delmarseafoods.com ·Email is the preferred channel; please keep all three parties copied.

1 目的

本文件汇总了 Del Mar Seafoods 在加利福尼亚州沃森维尔立式平板冻结项目中目前已确定的全部事项，并说明 Del Mar、狼王科技（Wolfking）与 APCCO 三方的分工。它是狼王开展设计工作的起点。本文件不取代此前邮件中已达成的任何约定，只是把它们集中在一处。

每项要求均标注状态，以明确责任方：

已确定 由 Del Mar 决定；除非 Del Mar 更改，否则固定不变 **狼王** 由狼王设计或提出方案

APCCO 由 APCCO 负责或确认 **待定** 尚未决定；责任方和时间见第 12 节

工作方式

所有项目沟通使用同一邮件线程（Joe 于 2026 年 10 月 30 日发起的 "Del Mar plate freezing projects"），每封邮件均抄送 Del Mar、狼王和 APCCO 三方。APCCO 的工程师会提出许多细节问题，这些问题是为了办理加州许可证，并不代表出了问题，请直接作答。

2 各方与职责

单位	在本项目中的职责
Del Mar Seafoods, Inc. 加州沃森维尔 — 业主与使用方	采购设备，拥有厂址和生产工艺。Joe Cappuccio（总裁）：商务决策、产品与进度。Kevin Cappuccio：项目工程、电气、技术对接。冻块尺寸、平面布置和最终设计在投产制造前须经 Del Mar 批准。
江苏狼王智能科技股份有限公司 江苏南通 — 设计与制造方	设计并制造 4 台立式平板冻结机（含液压站和控制系统），以及配套的整套氨制冷系统（优先考虑撬装/集装箱式机房，见第 6 节）。提供办理加州许可证所需的设计图纸、计算书、证书和技术文件。派 2 名技术人员到现场指导 APCCO 施工队安装，并主持调试。与合作伙伴一起支持 Del Mar 的第二阶段自动化（自动装料、卸料、装袋和装箱）。
APCCO — Applied Process Cooling Corp. 加州弗雷斯诺 — 持牌制冷承包商兼工程公司	审核狼王设计是否符合加州法规，编制并盖章许可证图纸和计算书，向沃森维尔市政府及其他主管部门申报许可证。安装全部到货设备：吊装就位、氨管道及保温、电气接线、控制系统集成。与狼王共同开机，并负责后续的维护和服务。APCCO 不制造冻结机，是 Del Mar 在当地的安装与合规合作方，将直接与狼王的工程师对接。

证书清单，事先约定

为避免狼王此前出口项目中遇到的问题（设备发运后在港口被要求补交文件），APCCO 将在投产制造前向狼王提供一份书面清单，列明本项目安装所需的全部证书、试验报告和文件；狼王确认哪些可以提供。 APCCO

3 产品与产能目标

同一批冻结机加工两种产品 已确定

- **整条加州鱿鱼**（*Doryteuthis opalescens*，加州市场鱿鱼），平板冻结成块。这是 Del Mar 面向中国市场的试销产品：如果中国客户接受平板冻结鱿鱼，Del Mar 将把更多产量转为这种方式。蒙特雷湾鱿鱼捕捞季 5 月开始，因此生产线必须在 2027 年 5 月前投入运行。
- **宠物食品原料**：鱿鱼绞碎后的糜状物，用同样的冻块冻结。该产品依据现有合同全年生产，是本项目的基础负荷，也是设备投资的回收来源。

日产能 已确定

45 t	2 × 10 h	4	8	140
每天（50 美吨），宠物食品合同	每天两班，另加 4 小时清洗	台立式平板冻结机，规格相同	每台每天批次（每班 4 批）	每批 10 kg 冻块数（目标）

Del Mar 的周期假设：冻结 90 分钟，加卸料和再装料 30 分钟，即每 2 小时一批，每班 4 批。45 t ÷ 8 批 ÷ 4 台 ≈ 每台每批 1.4 t，即 140 块 10 kg 冻块。140 块是目标；如果设计达不到，少一些也可以接受，但请尽量争取。

背景与后续计划

目前 Del Mar 用鼓风冻结：鱿鱼装入纸箱、上架、进冻结间，用工量很大，而加州的人工成本很高。本项目的目标是改为平板块冻，大幅减少人工搬运。这 4 台冻结机是试点：如果产品和经济性可行，Del Mar 将按同一设计成倍增加冻结机，因此请狼王按可复制的标准化设计考虑。第二阶段（计划 2027–2028 年）是自动化：冻结机自动装料、自动卸料、装袋、装箱并配检重秤。第一阶段的冻结机必须预留后期加装自动化的条件（见第 5 节）。

4 冻块规格

项目	要求	状态
冻块重量	标称 10 kg。鱿鱼密度比水低约 20%，因此 12 升的冻块约合 10 kg（10 kg 水 = 10 L）。	已确定

项目	要求	状态
冻块体积	标称 12,000,000 mm³（12.0 L）。	已确定
厚度	75 mm。固定值；厚度决定冻结时间。	已确定
冻块平面尺寸	目标约 535 × 300 mm（535 × 300 × 75 = 12.04 L）。Del Mar 对具体平面尺寸可以灵活，但一经选定即成为 Del Mar 今后所有冻结机的标准冻块尺寸。请狼王结合自身冻结板型材规格，提出既匹配板材又能保证 12 L 的平面尺寸；在订购冻结板之前须经 Del Mar 批准。	狼王
每格冻块数	青岛会面时的工作假设：每个板间（冻结格）设两块隔板，并排三块冻块。请狼王结合选定的平面尺寸、冻结板高度和冻块放置方向（哪一边竖直）予以确认。	狼王
每批冻块数	目标为每台每批 140 块（展会上讨论的是 117–120 块，Joe 在 10 月 30 日的邮件中提高到 140 块）。请尽量按接近 140 块设计，并说明相应的冻结板数量和机架长度。	狼王
冻结时间	Del Mar 假设 75 mm 冻块需 90 分钟。请狼王确认在设计蒸发温度下，整条鱿鱼和鱿鱼糜两种产品中心温度降至 −18 °C 所需的时间，并说明假设条件（进料温度、板温、接触状况）。	狼王
游离液体的排出	整条鱿鱼在装料和冻结初期会析出大量水分和墨汁。每个冻结格的底部必须能让这些游离液体排出，而不是封在格内，这样客户拿到的是一块鱿鱼，而不是一块冰。但宠物食品糜不能从格底流失。请狼王提出同时满足两种产品的底部设计（例如允许稀液通过但能留住糜料的排液缝/排液孔，或可调节的结构）。如有需要，Del Mar 可提供产品照片或样品。	狼王

5 立式平板冻结机要求

项目	要求	状态
数量与形式	4 台规格相同的立式平板冻结机，顶部装料，安装在室内加工车间。液压合板/开板，热氨脱模，液压顶升冻块。冻块从冻结机一侧卸到工作台或输送机上；卸料侧在平面布置中确定（第 8 节）。	已确定
产能	按第 3、4 节：75 mm 冻块，每格 3 块，目标每批 140 块，每天 8 批。	已确定
第一阶段装料	人工或半自动。Del Mar 正在考虑整条鱿鱼用顶部带闸门的输送机、鱿鱼糜用泵和软管灌装；欢迎狼王就 140 个冻结格最简单可靠的装料方式提出建议。最终由 Del Mar 决定。	待定

项目	要求	状态
预留第二阶段自动化	冻结机今后须能加装狼王的自动装料和卸料系统，而无需更换冻结机。请现在就把接口设计进去：顶部净空、装料龙门架的安装点或轨道、冻块卸料输送机接口以及备用控制 I/O。请狼王说明需要预先做好哪些预留。	狼王
卸料与包装	冻块顶升后进入装袋和装箱工序（冻块装袋，袋装箱），第二阶段计划增加检重秤。狼王曾提到合作伙伴的装袋和封箱设备，请作为单独报价的选项列入。	狼王
材料与卫生	冻结板：挤压铝合金，食品接触级。产品区和飞溅区（包括冻结板以上的所有部位）：不锈钢或铝，便于清洗，不得有镀锌表面。产品区以下的结构底座：请狼王提出不锈钢或热浸镀锌两种方案及差价；本厂每天接触海水和含氯消毒剂。	狼王
液压站	电机 460 V / 60 Hz（第 7 节）。安装在低温潮湿区域之外。液压油和密封件须与 NSF H1 食品级液压油兼容，油缸和软管处须耐 −35 °C。请狼王说明电机功率、工作压力，以及建议每台冻结机单独配液压站还是多台共用。	狼王
制冷剂连接	与活动冻结板连接的软管：不锈钢，氨用级，可从外部更换。氨系统任何部位不得使用铜或黄铜。	已确定
安全	急停、顶部开口处夹点防护、可上锁的电源隔离开关、液压能量隔离和泄压措施。防护按美国惯例，由 APCCO 审核（第 9 节）。	APCCO
技术文件	英文版：总装图、P&ID（管道仪表流程图）、电气原理图、操作维护手册、备件清单及推荐的两年备件包（软管、密封件、阀门）。冻结机和撬块的三维模型（STEP 格式），供 Del Mar 做平面布置。	已确定

6 制冷系统

由狼王设计并供应 4 台冻结机配套的整套制冷系统。APCCO 负责按加州法规审核、安装和维护。

项目	要求	状态
制冷剂	无水氨，R717。加州已不允许在此规模的新建系统中使用高 GWP 的氢氟烃（氟利昂）制冷剂，因此氨是实际可行的选择。	已确定
供货范围	压缩机组、冷凝器、容器、氨泵（如采用泵循环设计）、阀站、安全泄压装置、控制系统及控制柜，以及连接管道的设计。现场管道、保温和电气由 APCCO 安装。	已确定

项目	要求	状态
布置形式	Del Mar 在青岛会面时表达的倾向：采用撬装式或集装箱式机房，布置在室外紧邻厂房处，冻结机在室内，两者之间用保温管道连接。这样可节省厂房空间，并把压缩机噪音留在室外。Del Mar 与 APCCO 将于 11 月上旬在沃森维尔现场开会，确定机房位置和管道走向；请狼王在此之前提供标准撬装机房的尺寸和概念布置图。Joe 还询问过的替代方案：每台冻结机配一套小型独立制冷机组；请狼王就可行性、每套充注量以及与一套集中机房相比的成本发表意见。	待定
氨充注量	最小充注量是设计目标之一。加州 CalARP（事故性泄漏预防）计划的氨触发量为现场 500 磅（227 kg），低于此值可大幅简化许可程序。请狼王计算所提设计方案的总充注量。Del Mar 理解在此产能下可能无法达到；如果做不到，请说明充注量，以便 APCCO 安排相应申报。	狼王
设计工况	APCCO 在每批冻块数提高到 140 块之前的初步估算：4 台冻结机约 45 冷吨（158 kW），蒸发温度 −30 °F（−34 °C），按泵循环、循环倍率 4:1 假设。请狼王按最终每批冻块数自行计算负荷（含批次错开时的峰值降温负荷），并给出相应的压缩机和冷凝器选型。	狼王
系统形式	请狼王说明：泵循环供液或其他供液方式及循环倍率；蒸发温度和冷凝温度；冷凝器形式（蒸发式或风冷式；本现场对用水和噪音都有要求）；带贮液器还是临界充注设计；融霜方式（假定热氨融霜）及冻结机对热氨的要求。	狼王
阀门与控制	请狼王提供阀门和仪表设备清单（含品牌），并说明哪些阀门随冻结机装配发货、哪些散件发货、哪些由 APCCO 供应。说明冻结机控制柜是否控制制冷阀门。APCCO 将确认哪些品牌在加州便于维修服务。	狼王
压力设计	压力容器按 ASME 第 VIII 卷第 1 册制造，带 U 钢印并在美国国家锅炉压力容器检验师委员会（National Board）注册。管道按 ASME B31.5。设计压力按 IIAR 2；热氨融霜时冻结板及其管路承受排气侧压力，因此必须按该压力设计（具体数值由 APCCO 确认，通常为 250 psig / 17.2 bar）。按加州压力容器法规要求对容器焊缝进行射线检测。材料质保书和焊工资质应能应要求提供。	APCCO

7 电气

项目	要求	状态
现场电源	480 V，三相，60 Hz（480Y/277 V，星形接地）。所有电机铭牌按 460 V / 60 Hz。美国没有 380 V 电源，也没有 220 V 单相电源。	已确定
控制电源	120 VAC，60 Hz，单相，由每个控制柜内的控制变压器提供；仪表、I/O 和电磁阀采用 24 VDC。不得使用任何 220 V 单相元件（加热器、风扇、照明、线圈）。	已确定
控制柜	按 UL 508A 标准制造和标识（取得 UL 或 ETL 的 UL 508A 认证），标注短路电流额定值（SCCR）。冲洗区域采用 NEMA 4X 不锈钢柜体。元器件须有 UL 认证（Listed 或 Recognized）：施耐德、ABB、西门子、伊顿、Allen-Bradley 或同等品牌。接线工艺和线色按 NFPA 79；标识用英文。	已确定
电机	NEMA Premium / IE3 能效等级。美国能源部规定进口电机即使装在整机内也必须满足该能效要求。液压泵和风机电机须按 60 Hz 转速选型，不得用 50 Hz 电机换铭牌。	已确定
负荷清单，请优先提供	按每台冻结机和每个撬块列出：功率 kW、480 V 下的满载电流、启动方式和启动电流，以及建议的馈线数量和规格。Del Mar 需要凭此向电力公司（PG&E）申请用电，该流程可能比设备制造周期更长。	狼王
控制系统	PLC 和触摸屏采用英文界面。欢迎狼王远程接入提供支持，Del Mar 提供网络连接。请狼王提出 PLC 品牌建议；考虑到在美国的售后服务，优先西门子或 Allen-Bradley。	狼王

8 现场：加州沃森维尔

- **地点。**Del Mar Seafoods，加利福尼亚州沃森维尔，位于蒙特雷湾，旧金山以南约 150 km。沿海气候温和。冷凝器的设计环境温度由 APCCO 提供（夏季设计工况大约为干球 29 °C / 湿球 18 °C，待确认）。APCCO
- **平面布置。**冻结机安装在室内加工车间；撬装机房布置在室外紧邻厂房；管道走向和长度在 11 月上旬 Del Mar 与 APCCO 的现场会议上确定。请狼王说明机房与冻结机之间建议的最大距离。待定
- **抗震。**沃森维尔处于高烈度地震区（1989 年洛马普列塔地震震中就在附近）。所有设备都需要按加州地震荷载设计的锚固。请狼王提供冻结机、撬块和容器的重量、重心和地脚螺栓位置；锚固计算由 APCCO 的工程师完成。狼王
- **地面与基础。**请狼王提供冻结机和撬装机房的占地尺寸和集中荷载。狼王

- **现有公用设施。**480 V 三相电源、饮用水、排水、网络。压缩空气待确认。
- **运输。**海运至奥克兰港或长滩港，再陆运至沃森维尔。撬装机房应符合标准集装箱尺寸或合法公路运输尺寸；现场吊装通道由 APCCO 提供建议。贸易条款在报价中商定。
  待定

9 设计阶段需要注意的法规要点

只列出会影响狼王设计内容的要点。所有这些事项均以 APCCO 为准，具体数值由 APCCO 提供；在此列出是为了避免意外。

1. **许可证。**沃森维尔市政府依据由加州注册专业工程师（APCCO 的 P.E.）盖章的图纸和计算书签发建筑、机械和电气许可证。狼王提供设计，APCCO 负责调整、盖章和申报。许可证办理时间可能长于设备制造周期，因此图纸要先行。
2. **氨系统设计。**ANSI/IIAR 2-2021（闭式氨制冷系统设计标准），以及 IIAR 4、5、6（安装、开机、检验），还有《加州机械规范》制冷章节。这些标准规定了设计压力、安全阀、阀门布置和标识、紧急停机，以及室内机房的氨泄漏检测和通风要求。采用室外撬装机房可减少室内方面的要求。
3. **氨存量触发值。**加州 CalARP 计划自 500 磅（227 kg）起适用；联邦 OSHA PSM 和 EPA RMP 自 10,000 磅起适用。按最小充注量设计，并书面记录充注量。
4. **容器与管道。**容器按 ASME 第 VIII 卷第 1 册带 U 钢印并在 National Board 注册；制冷管道按 ASME B31.5；需办理加州 Cal/OSHA 压力容器许可证，按要求对容器焊缝做射线检测。材料质保书（MTR）和按 ASME 第 IX 卷的焊工资质须能提供。
5. **电气。**《加州电气规范》（NEC）。所有电气设备必须取得美国认可检测机构（UL、ETL、CSA）的认证：控制柜按 UL 508A，电机和元器件有 Listed 或 Recognized 认证，电机满足美国能源部能效要求。未认证的设备需要现场评估，既慢又贵，请在设计阶段避免。
6. **制冷剂。**加州空气资源委员会（CARB）禁止在充注量超过 50 磅的新建制冷系统中使用高 GWP 氢氟烃；氨是合规选择，已经确定。
7. **抗震。**《加州建筑规范》和 ASCE 7：设备锚固和限位须按现场条件设计。撬块、容器和冻结机机架需要设计好的锚固点。
8. **食品安全。**美国 FDA《食品安全现代化法案》（FSMA）：水产品 HACCP（21 CFR 123）和动物食品预防性控制（21 CFR 507）。产品接触面必须耐腐蚀、易清洗；可能接触产品的润滑剂必须是 NSF H1 食品级；产品上方不得有会剥落的涂层。
9. **人员安全。**加州 Cal/OSHA：机械防护、急停、上锁挂牌（LOTO）条件、英文标识和警示。
10. **噪音。**厂房旁的室外设备在地界处必须满足市政噪音限值。这会影响冷凝器风机选型以及机房是否需要隔音处理。位置确定后，Del Mar 和 APCCO 将提供限值。

10 进度

时间	内容	责任方
2026 年 11 月 2 日当周	发布本简报。Del Mar 发送产品照片和样品。Del Mar 与 APCCO 在沃森维尔现场开会，确定机房位置和平面布置。狼王开始设计。	各方
尽快	初步总装图、P&ID、电气负荷清单、氨充注量估算和报价（第 11 节）。Del Mar 和 APCCO 审核；基于初步设计启动许可证工作。	狼王
设计批准后	冻结机和制冷系统制造。狼王在展会上的估计：约 100 天。	狼王
运输与安装	海运、清关、吊装、安装、调试。安装工期和人员由 APCCO 确认。	APCCO + 狼王技术人员
2027 年 5 月	目标：在鱿鱼捕捞季开始前完成安装和调试。最迟可接受的时间是 2027 年 6 月。Joe 已告知宠物食品客户总周期为六个月。	各方
2027–2028 年	第二阶段：自动装料和卸料、装袋和装箱、检重。	狼王 + 合作伙伴

11 首先需要狼王提供的内容

1. **单台冻结机初步总装图：**外形尺寸、冻结板数量、冻结格布置、建议的冻块平面尺寸、装料高度、卸料侧、重量。
2. **冻结时间计算：**75 mm 整条鱿鱼冻块和鱿鱼糜冻块，附假设条件。
3. **制冷系统方案：**系统形式、P&ID、设备清单、氨充注量估算、撬装机房概念方案（含尺寸和重量）。
4. **电气负荷清单**和建议的单线图；控制柜规格及认证方式（UL 或 ETL）。
5. **容器清单**（含 ASME 和 National Board 状态）；焊接和质量文件计划；狼王可提供的证书清单（与 APCCO 的清单对照，见第 2 节）。
6. **商务报价：**狼王供货与 APCCO 安装的范围划分、价格、交期、付款条件、质保、推荐备件包，以及现场技术人员的条件（人数、天数、差旅）。
7. **三维模型**（STEP 格式）：冻结机和撬装机房，供 Del Mar 做平面布置，有了即发。
8. **业绩清单：**在美国的安装项目，包括青岛会面时谈到的俄勒冈和洛杉矶客户。
9. **第二阶段预算概念：**自动装料和卸料，以及合作伙伴的装袋和装箱设备，以便现在就把接口设计进冻结机。

12 待定事项

事项	责任方	需要时间
最终冻块平面尺寸（约 535 × 300 × 75 mm，12 L）	狼王提出，Del Mar 批准	订购冻结板之前
每批冻块数（目标 140）和冻结板数量	狼王	随初步总装图
冻结时间确认	狼王	随初步总装图
冻结格底部设计：排出整鱿鱼析出液，留住糜料	狼王	随初步总装图
机架底座材料（不锈钢或热浸镀锌）及差价	狼王	随报价
每台单独液压站还是共用	狼王	随初步总装图
机房：集装箱式还是开放式撬块、位置、隔音处理	Del Mar + APCCO 现场会议	11 月上旬
氨充注量及 CalARP 适用情况	狼王计算，APCCO 审核	随制冷方案
冷凝器形式及供水条件	狼王提出，APCCO 确认	随制冷方案
热氨融霜要求及阀门供货范围划分	狼王 + APCCO	随 P&ID
用于 PG&E 用电申请的电气负荷清单	狼王	尽快
第一阶段装料方式	Del Mar 决定，狼王建议	随总装图
贸易条款和港口	Del Mar + 狼王	随报价
安装工期和人员规模	APCCO	初步设计之后

13 联系人

Del Mar Seafoods

Joe Cappuccio
总裁
joe@delmarseafoods.com
+1 831-760-3005

Del Mar Seafoods

Kevin Cappuccio
项目工程、电气
arabidsquid@gmail.com

APCCO

Alex Robinson
施工部门经理
arobinson@apcco.net
+1 559-875-7720 分机 411 • 手机 +1 559-696-6227

APCCO
Patrick Cocagne, P.E.
工程副总裁
pcocagne@apcco.net
+1 559-875-7720 分机 402 · 手机 +1 559-358-5063

狼王科技
Sean Zhu
江苏狼王智能科技股份有限公司
sean@wolfkingtech.com

狼王科技
David Ye
江苏狼王智能科技股份有限公司
david@wolfkingtech.com

Del Mar Seafoods, Inc. · 美国加州沃森维尔 · www.delmarseafoods.com · 请以电子邮件为主要沟通方式，并始终抄送三方。